

Homework 6: The Rotating Wave Approximation



In the lecture notes we studied Rabi oscillations driven by a circularly polarized field, for which the problem could be solved exactly. In practice, however, the driving field is often *linearly* polarized. In this exercise you will show that the linearly polarized problem reduces to the circularly polarized one after an approximation known as the *rotating wave approximation* (RWA), and you will estimate the leading correction due to the terms that are dropped. Throughout, we work in natural units with $\hbar = 1$ and use the free Hamiltonian

$$\hat{H}_0 = \frac{\omega_0}{2} \sigma_z, \quad (1)$$

with eigenstates $|+\rangle$ (energy $+\omega_0/2$) and $|-\rangle$ (energy $-\omega_0/2$), and a *linearly polarized* perturbation

$$\hat{V}(t) = \Omega \cos(\omega t) \sigma_x, \quad (2)$$

where $\Omega \ll \omega_0$ and the driving frequency ω is near resonance, $|\omega - \omega_0| \ll \omega_0$.

(a) Decomposition into rotating components. Using $\sigma_x = \sigma_+ + \sigma_-$ with $\sigma_{\pm} = \frac{1}{2}(\sigma_x \pm i\sigma_y)$, show that $\hat{V}(t)$ can be written as

$$\hat{V}(t) = \hat{V}^{\text{co}}(t) + \hat{V}^{\text{ctr}}(t), \quad (3)$$

where

$$\hat{V}^{\text{co}}(t) = \frac{\Omega}{2}(\sigma_+ e^{-i\omega t} + \sigma_- e^{+i\omega t}), \quad \hat{V}^{\text{ctr}}(t) = \frac{\Omega}{2}(\sigma_+ e^{+i\omega t} + \sigma_- e^{-i\omega t}). \quad (4)$$

Compare $\hat{V}^{\text{co}}(t)$ with the perturbation used in the lecture notes (Eq. 3 of the Rabi notes). What is the physical interpretation of these two terms?

(*Hint: Consider which component of a classical magnetic field rotates in the same sense as the Larmor precession.*)

(b) Interaction picture. Go to the interaction picture. Using $e^{i\hat{H}_0 t} \sigma_{\pm} e^{-i\hat{H}_0 t} = \sigma_{\pm} e^{\pm i\omega_0 t}$, show that

$$\hat{V}_I(t) = \frac{\Omega}{2} \begin{pmatrix} 0 & e^{-i\delta t} + e^{i(\omega_0 + \omega)t} \\ e^{i\delta t} + e^{-i(\omega_0 + \omega)t} & 0 \end{pmatrix}, \quad (5)$$

where $\delta = \omega - \omega_0$ is the detuning. Identify which terms in this matrix originate from $\hat{V}^{\text{co}}$ and which from $\hat{V}^{\text{ctr}}$. Near resonance ($\delta \approx 0$), argue that the co-rotating terms vary slowly (at frequency $|\delta|$) while the counter-rotating terms oscillate rapidly (at frequency $\omega_0 + \omega \approx 2\omega_0$).

(c) The rotating wave approximation. The RWA consists of dropping the rapidly oscillating counter-rotating terms, i.e., replacing $\hat{V}_I(t) \rightarrow \hat{V}_I^{\text{RWA}}(t)$ with

$$\hat{V}_I^{\text{RWA}}(t) = \frac{\Omega}{2} \begin{pmatrix} 0 & e^{-i\delta t} \\ e^{i\delta t} & 0 \end{pmatrix}. \quad (6)$$

Observe that this is identical to the interaction picture potential from the circularly polarized Rabi problem (Eq. 6 of the lecture notes). Conclude that, within the RWA, the transition probability for a linearly polarized driver is given by

$$P_{+\rightarrow-}^{\text{RWA}}(t) = \frac{\Omega^2}{\delta^2 + \Omega^2} \sin^2\left(\frac{\sqrt{\delta^2 + \Omega^2} t}{2}\right). \quad (7)$$

Under what conditions is the RWA expected to be a good approximation? Express your answer as an inequality involving Ω , ω , and ω_0 .

(d) First-order correction from the counter-rotating terms. Let us now estimate the error introduced by the RWA using first-order perturbation theory. Compute the first-order transition amplitude $|+\rangle \rightarrow |-\rangle$ arising from the counter-rotating part alone:

$$A_{+\rightarrow-}^{\text{ctr}}(t) = -i \int_0^t dt' \langle - | \hat{V}_I^{\text{ctr}}(t') | + \rangle. \quad (8)$$

Show that, near resonance, their ratio scales as

$$\frac{|A^{\text{ctr}}|}{|A^{\text{co}}|} \sim \frac{|\delta|}{\omega_0 + \omega} \approx \frac{|\delta|}{2\omega_0}, \quad (9)$$

confirming that the counter-rotating contribution is negligible when $|\delta| \ll \omega_0$.

This question is completely optional but will count 1 full point towards your exam score.

(Bonus) The Bloch-Siegert shift Although the counter-rotating terms do not contribute to the $|+\rangle \rightarrow |-\rangle$ transition at first order in any significant way, they *do* affect the diagonal matrix elements at second order, producing a shift in the effective resonance condition. This is called the Bloch-Siegert shift.

Consider the second-order correction to the coefficient of the initial state $|+\rangle$. Using the Dyson series, this is

$$c_+^{(2)}(t) = (-i)^2 \int_0^t dt' \int_0^{t'} dt'' \langle + | \hat{V}_I^{\text{ctr}}(t') \hat{V}_I^{\text{ctr}}(t'') | + \rangle. \quad (10)$$

Note that we use only $\hat{V}_I^{\text{ctr}}$, since the cross terms between co- and counter-rotating pieces and the purely co-rotating contribution are already accounted for in the RWA solution. Indeed, the cross terms between them in $c_+^{(2)}$ would involve intermediate state $|-\rangle$ with one co-rotating and one counter-rotating vertex; these carry oscillatory phases $e^{\pm i\delta t'} e^{\pm i\Sigma t''}$.

- (i) Show that $\langle + | \hat{V}_I^{\text{ctr}}(t') \hat{V}_I^{\text{ctr}}(t'') | + \rangle = \frac{\Omega^2}{4} e^{i(\omega_0 + \omega)(t' - t'')}$.

(Hint: The product $\hat{V}_I^{\text{ctr}}(t') \hat{V}_I^{\text{ctr}}(t'')$ has a diagonal part proportional to $\sigma_+ \sigma_- + \sigma_- \sigma_+$. Only one of these contributes when sandwiched between $\langle + | \dots | + \rangle$.)

- (ii) Perform the double time integral and show that, for $(\omega_0 + \omega)t \gg 1$, the dominant contribution takes the form

$$c_+^{(2)}(t) \approx -\frac{i\Omega^2}{4(\omega_0 + \omega)} t. \quad (11)$$

(Hint: After performing the t'' integral, you will encounter a term that grows linearly in t and oscillatory terms that remain bounded. Keep only the secular (linearly growing) piece.)

- (iii) Interpret this result. Argue that the counter-rotating terms effectively shift the energy of $|+\rangle$ by $\Delta E_+ = +\frac{\Omega^2}{4(\omega_0 + \omega)}$. By a similar calculation (or by symmetry), argue that $|-\rangle$ is shifted by $\Delta E_- = -\frac{\Omega^2}{4(\omega_0 + \omega)}$.

- (iv) Conclude that the resonance condition is shifted from $\omega = \omega_0$ to

$$\omega = \omega_0 + \frac{\Omega^2}{2(\omega_0 + \omega)} \approx \omega_0 + \frac{\Omega^2}{4\omega_0}, \quad (12)$$

where the last approximation holds near resonance. This is the Bloch-Siegert shift. Estimate its magnitude relative to ω_0 for typical atomic physics parameters ($\omega_0 \sim 10^{15}$ rad/s, $\Omega \sim 10^9$ rad/s) and comment on when it might become experimentally relevant.

Solutions



HW 6.1: the problem guides you through the solution step by step, but I have added some additional comments and discussion along the way.

(a) We write $\cos(\omega t) = \frac{1}{2}(e^{-i\omega t} + e^{+i\omega t})$ and $\sigma_x = \sigma_+ + \sigma_-$, so that

$$\hat{V}(t) = \Omega \cos(\omega t) \sigma_x = \frac{\Omega}{2}(e^{-i\omega t} + e^{+i\omega t})(\sigma_+ + \sigma_-) \quad (13)$$

$$= \frac{\Omega}{2}(\sigma_+ e^{-i\omega t} + \sigma_- e^{+i\omega t}) + \frac{\Omega}{2}(\sigma_+ e^{+i\omega t} + \sigma_- e^{-i\omega t}) \quad (14)$$

$$\equiv \hat{V}^{\text{co}}(t) + \hat{V}^{\text{ctr}}(t). \quad (15)$$

Comparing $\hat{V}^{\text{co}}(t) = \frac{\Omega}{2}(\sigma_+ e^{-i\omega t} + \sigma_- e^{+i\omega t})$ with Eq. 3 of our Rabi notes, we see that it is identical to the circularly polarized perturbation studied there.

Physically, the linearly polarized oscillation $\cos(\omega t) \sigma_x$ decomposes into two circular polarizations. In a spin- $\frac{1}{2}$ system, the Larmor precession occurs at frequency ω_0 . Recall that when setting $\Omega \rightarrow 0$, we had

$$|\psi(t)\rangle = e^{-i\hat{H}_0 t} |\psi(0)\rangle = \begin{pmatrix} e^{-i\omega_0 t/2} & 0 \\ 0 & e^{+i\omega_0 t/2} \end{pmatrix} |\psi(0)\rangle, \quad (16)$$

with the phase evolution of $|+\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|-\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ having the same sign structure as the co-rotating piece of the Ω Hamiltonian. You should interpret this rotation as the spin precessing in the x - y plane around the z axis. The co-rotating component $\hat{V}^{\text{co}}$ rotates in the same “direction” as this precession. The intuition here is that it advances the phase of the spin and can therefore efficiently drive transitions. The counter-rotating component $\hat{V}^{\text{ctr}}$ rotates in the opposite direction and is “out of sync” with the precession and its effect averages out over time.

(b) Interaction picture. The interaction picture potential is $\hat{V}_I(t) = e^{i\hat{H}_0 t} \hat{V}(t) e^{-i\hat{H}_0 t}$, with $\hat{H}_0 = \frac{\omega_0}{2} \sigma_z$. Using the provided identity $e^{i\hat{H}_0 t} \sigma_{\pm} e^{-i\hat{H}_0 t} = \sigma_{\pm} e^{\pm i\omega_0 t}$, which follows from $[\sigma_z, \sigma_{\pm}] = \pm 2\sigma_{\pm}$, we have

$$\hat{V}_I(t) = \frac{\Omega}{2} [\sigma_+ e^{i\omega_0 t} (e^{-i\omega t} + e^{+i\omega t}) + \sigma_- e^{-i\omega_0 t} (e^{-i\omega t} + e^{+i\omega t})] \quad (17)$$

$$= \frac{\Omega}{2} [\sigma_+ (e^{-i(\omega-\omega_0)t} + e^{i(\omega+\omega_0)t}) + \sigma_- (e^{i(\omega-\omega_0)t} + e^{-i(\omega+\omega_0)t})]. \quad (18)$$

Using $\omega - \omega_0 = \delta$ and writing $\sigma_+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $\sigma_- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, we obtain

$$\hat{V}_I(t) = \frac{\Omega}{2} \begin{pmatrix} 0 & e^{-i\delta t} + e^{i(\omega_0+\omega)t} \\ e^{i\delta t} + e^{-i(\omega_0+\omega)t} & 0 \end{pmatrix}. \quad (19)$$

The co-rotating terms ($e^{\pm i\delta t}$) originate from $\hat{V}^{\text{co}}$, and the counter-rotating terms ($e^{\pm i(\omega_0+\omega)t}$) originate from $\hat{V}^{\text{ctr}}$. Near resonance, $\delta \simeq 0$, so the co-rotating terms vary slowly (at frequency $|\delta| \ll \omega_0$), while the counter-rotating terms oscillate rapidly at frequency $\omega_0 + \omega \approx 2\omega_0$.



The key observation of the RWA is that if $\delta = \omega - \omega_0$ is small, the co-rotating terms add no additional fast oscillations beyond the free evolution, while the counter-rotating terms add a very fast oscillation at frequency $\omega + \omega_0$.

(c) Dropping the counter-rotating terms gives

$$\hat{V}_I^{\text{RWA}}(t) = \frac{\Omega}{2} \begin{pmatrix} 0 & e^{-i\delta t} \\ e^{i\delta t} & 0 \end{pmatrix}, \quad (20)$$

which is identical to Eq. 6 of the corrected Rabi notes. Since the two problems now have the same interaction-picture Hamiltonian, they have the same time evolution, and we can immediately import the exact result:

$$P_{+\rightarrow-}^{\text{RWA}}(t) = \frac{\Omega^2}{\delta^2 + \Omega^2} \sin^2\left(\frac{\sqrt{\delta^2 + \Omega^2} t}{2}\right). \quad (21)$$

The RWA is valid when the counter-rotating terms oscillate so rapidly that they average out before having any appreciable effect. The relevant comparison is the strength of the perturbation Ω (which sets the timescale $\sim 1/\Omega$ over which transitions develop) against the frequency of the counter-rotating oscillations $\omega + \omega_0$. The RWA is therefore a good approximation when

$$\Omega \ll \omega + \omega_0. \quad (22)$$

Near resonance $\omega \approx \omega_0$, this reduces to $\Omega \ll 2\omega_0$, which is simply the requirement that the driving is weak compared to the level spacing.

(d) First-order correction from the counter-rotating terms. The counter-rotating part of $\hat{V}_I$ has matrix elements

$$\langle - | \hat{V}_I^{\text{ctr}}(t') | + \rangle = \frac{\Omega}{2} e^{-i(\omega_0 + \omega)t'}, \quad (23)$$

reading off the (2,1) entry. The first-order amplitude is therefore

$$A_{+\rightarrow-}^{\text{ctr}}(t) = -i \int_0^t dt' \frac{\Omega}{2} e^{-i(\omega_0 + \omega)t'} = -i \frac{\Omega}{2} \frac{e^{-i(\omega_0 + \omega)t} - 1}{-i(\omega_0 + \omega)} \quad (24)$$

$$= -\frac{\Omega}{2} \frac{e^{-i(\omega_0 + \omega)t} - 1}{\omega_0 + \omega}. \quad (25)$$

For the co-rotating part, the (2,1) entry is $\frac{\Omega}{2} e^{i\delta t'}$, and an identical calculation gives

$$A_{+\rightarrow-}^{\text{co}}(t) = -\frac{\Omega}{2} \frac{e^{i\delta t} - 1}{\delta}. \quad (26)$$

Taking magnitudes using $|e^{i\alpha} - 1| = 2|\sin(\alpha/2)|$:

$$|A^{\text{ctr}}| = \frac{\Omega}{\omega_0 + \omega} \left| \sin\left(\frac{(\omega_0 + \omega)t}{2}\right) \right|, \quad |A^{\text{co}}| = \frac{\Omega}{|\delta|} \left| \sin\left(\frac{\delta t}{2}\right) \right|. \quad (27)$$

The ratio of the sine functions is bounded and of order unity on average (both oscillate between 0 and 1), so the ratio of amplitudes scales as

$$\frac{|A^{\text{ctr}}|}{|A^{\text{co}}|} \sim \frac{|\delta|}{\omega_0 + \omega} \approx \frac{|\delta|}{2\omega_0}. \quad (28)$$

Near resonance ($|\delta| \ll \omega_0$), this ratio is very small, confirming that the counter-rotating contribution to the transition amplitude is negligible.



HW 6.1 (Bonus): the Bloch-Siegert shift. This is a more subtle effect, but it is another great demonstration of perturbation theory.

(i) We need to evaluate the matrix element of the product $\hat{V}_I^{\text{ctr}}(t')\hat{V}_I^{\text{ctr}}(t'')$ between $\langle + | \dots | + \rangle$. Writing out the counter-rotating piece:

$$\hat{V}_I^{\text{ctr}}(t) = \frac{\Omega}{2}(\sigma_+ e^{i(\omega_0 + \omega)t} + \sigma_- e^{-i(\omega_0 + \omega)t}). \quad (29)$$

The product has four terms. Two of them vanish because $\sigma_+^2 = \sigma_-^2 = 0$. The surviving terms are

$$\hat{V}_I^{\text{ctr}}(t')\hat{V}_I^{\text{ctr}}(t'') = \frac{\Omega^2}{4}[\sigma_+\sigma_- e^{i(\omega_0 + \omega)(t' - t'')} + \sigma_-\sigma_+ e^{-i(\omega_0 + \omega)(t' - t'')}]. \quad (30)$$

Now, $\sigma_+\sigma_- = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $\sigma_-\sigma_+ = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$. Since $|+\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, only the $\sigma_+\sigma_-$ term survives:

$$\langle + | \hat{V}_I^{\text{ctr}}(t')\hat{V}_I^{\text{ctr}}(t'') | + \rangle = \frac{\Omega^2}{4} e^{i(\omega_0 + \omega)(t' - t'')}. \quad (31)$$

(ii) **Double time integral.** Defining $\Sigma \equiv \omega_0 + \omega$,

$$c_+^{(2)}(t) = -\frac{\Omega^2}{4} \int_0^t dt' \int_0^{t'} dt'' e^{i\Sigma(t' - t'')} = -\frac{\Omega^2}{4} \int_0^t dt' e^{i\Sigma t'} \int_0^{t'} dt'' e^{-i\Sigma t''}. \quad (32)$$

The inner integral is

$$\int_0^{t'} dt'' e^{-i\Sigma t''} = \frac{1 - e^{-i\Sigma t'}}{i\Sigma}. \quad (33)$$

Substituting:

$$c_+^{(2)}(t) = -\frac{\Omega^2}{4i\Sigma} \int_0^t dt' (e^{i\Sigma t'} - 1) = -\frac{\Omega^2}{4i\Sigma} \left[\frac{e^{i\Sigma t} - 1}{i\Sigma} - t \right]. \quad (34)$$

The first term in the brackets is oscillatory and bounded (its magnitude is at most $2/\Sigma$), while t grows without bound.



Terms that grow with time (as opposed to oscillating within a bounded range) are called secular terms.

For $\Sigma t \gg 1$, the secular piece dominates:

$$c_+^{(2)}(t) \approx -\frac{\Omega^2}{4i\Sigma} \cdot (-t) = \frac{\Omega^2 t}{4i\Sigma} = -\frac{i\Omega^2}{4(\omega_0 + \omega)} t. \quad (35)$$

(iii) **Interpretation as an energy shift.** The total coefficient of $|+\rangle$ through second order is

$$c_+(t) = 1 - \frac{i\Omega^2}{4(\omega_0 + \omega)} t + \mathcal{O}(V^4), \quad (36)$$

where we used that $c_+^{(1)}(t) = 0$ for the counter-rotating terms (the first-order diagonal correction vanishes because $\langle + | \hat{V}_I^{\text{ctr}}(t') | + \rangle = 0$ – both σ_+ and σ_- are off-diagonal). This has the form of the early-time expansion of a pure phase:

$$c_+(t) \approx e^{-i\Delta E_+ t}, \quad \Delta E_+ = +\frac{\Omega^2}{4(\omega_0 + \omega)}. \quad (37)$$

The counter-rotating terms therefore shift the energy of $|+\rangle$ upward by ΔE_+ .

For $|-\rangle$, the only change is that $\sigma_- \sigma_+$ now contributes instead of $\sigma_+ \sigma_-$, and the phase is conjugated:

$$\langle - | \hat{V}_I^{\text{ctr}}(t') \hat{V}_I^{\text{ctr}}(t'') | - \rangle = \frac{\Omega^2}{4} e^{-i\Sigma(t'-t'')}. \quad (38)$$

Solving analogous integrals, we have

$$c_-^{(2)}(t) = -\frac{\Omega^2}{4i\Sigma} \left[t + \frac{e^{-i\Sigma t} - 1}{i\Sigma} \right] \simeq -\frac{\Omega^2 t}{4i\Sigma} = +\frac{i\Omega^2}{4(\omega_0 + \omega)} t. \quad (39)$$

So $c_-(t) \simeq e^{-i\Delta E_- t}$ with

$$\Delta E_- = -\frac{\Omega^2}{4(\omega_0 + \omega)}. \quad (40)$$

The energy of $|-\rangle$ is shifted downward by the same amount.



A quicker way to arrive at this by noting that the Hamiltonian $\hat{V}^{\text{ctr}}$ has a symmetry under simultaneous exchange $|+\rangle \leftrightarrow |-\rangle$ and $\Sigma \rightarrow -\Sigma$ (equivalently, complex conjugation of the time-dependent phases), which reverses the sign of the shift.

(iv) The shifted resonance condition. The counter-rotating terms increase the energy splitting between $|+\rangle$ and $|-\rangle$:

$$\omega_0^{\text{eff}} = \omega_0 + \Delta E_+ - \Delta E_- = \omega_0 + \frac{\Omega^2}{2(\omega_0 + \omega)}. \quad (41)$$

The resonance condition $\omega = \omega_0^{\text{eff}}$ then becomes

$$\omega = \omega_0 + \frac{\Omega^2}{2(\omega_0 + \omega)}. \quad (42)$$

Near resonance ($\omega \simeq \omega_0$), we replace $\omega_0 + \omega \approx 2\omega_0$ on the right-hand side to obtain the Bloch-Siegert shift:

$$\boxed{\delta_{\text{BS}} \equiv \omega_{\text{res}} - \omega_0 \approx \frac{\Omega^2}{4\omega_0}.} \quad (43)$$

The shift is always positive: the resonance moves to higher driving frequency. Its relative magnitude is

$$\frac{\delta_{\text{BS}}}{\omega_0} \approx \frac{\Omega^2}{4\omega_0^2} = \frac{1}{4} \left(\frac{\Omega}{\omega_0} \right)^2. \quad (44)$$

For the quoted parameters ($\omega_0 \sim 10^{15}$ rad/s, $\Omega \sim 10^9$ rad/s):

$$\frac{\delta_{\text{BS}}}{\omega_0} \sim \frac{10^{18}}{4 \times 10^{30}} \sim 10^{-13}. \quad (45)$$

This is utterly negligible in atomic physics and optical experiments, which is why the RWA works so well in those settings.